Yiwen Chen

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EDUCATION

- **M.Sc., China University of Geosciences (Wuhan)** 2022.09-2025.06

Computer Science **GPA: 92.03125/100 Top 1%**

Relevant Courses: Machine Learning and Data Mining 92 Algorithm Design 90 Intelligent Computing92 Software Engineering 92 Computer Applied Mathematics 89

- **B.S., Beijing Language and Culture University** 2018.09-2022.06

Digital Media Technology **GPA:3.62/4.0 Top 10%**

Relevant Courses: C++ 92 Computer Networks 93 Computer Graphics 93 Digital Image Processing92 Database 88 Operating Systems 88 Virtual Reality 91 Digital Signal processing

PROFESSIONAL SKILLS

Abundant Experience in Deep Learning and AI:

- Familiar with deep learning-related knowledge, such as Generative AI, CNN, RL, LMM, RAG, Finetuning, etc,
- Proficient in using the Pytorch, understand Tensorflow, and have practiced some classic AI projects, such as Handwritten Digit Image Recognition, Kaggle classic project Cat and Dog Binary Classification, through PyTorch Geometric Library, using GCN method for 7 classification tasks on Cora dataset, etc.
- Be skillful in Python language, especially various Python packages such as Numpy, Pandas, Scikit-learn, etc.

Solid CS Basic Knowledge:

- Good at common Data Structures and Algorithms, Computer Networks, Operating Systems, MySQL relational databases, and Principles of Computer Composition, etc.
- Have a knowledge of using Linux operating system, and familiar with various Linux commands and shell script programming.

Good at Mathematics:

- Second Prize in the Beijing Division of the 12th National College Student Mathematics Competition (2020)
- Second prize in the Hubei Division of the National Mathematics/ Chemistry / Biology Olympiad(2017)

Programming: C/C++/Python/JAVA/C #/Matlab/CUDA, master Object-Oriented programming concepts.

Tools: Git/CMake/Gdb/Vim/Springboot/Qt/Unity/3D-Max

Data Analysis&Visualization: CSS/Javascript/Vue/Html/Echarts/SPSS

Good English Ability: IELTS 7/ Test for English Major-4

RESEARCH EXPERIENCE

Graph Convolutional Network for Water Network Partitioning 2023.09-2024.06

As First Author, Accepted, *Expert systems with applications* (**JCR Q1, IF=7.5**)

- Research project completed independently from 0 to 1.
- By reading a large amount of literature, I found it is essential to utilize the GCN algorithm to combine pressure data and structural topology information comprehensively to cluster the network nodes which nearly nobody has done before.
- Learning materials about deep learning and hydrology, I simulated and generated datasets with Python code.
- On the PyTorch platform, I tried and conducted various experiments about different GCN methods such as Structural Deep Clustering Networks and KMeans on different water networks.
- By analyzing the result, I found a Cross-Attention Fusion Based Enhanced Graph Convolutional Network for Clustering performance well, thus I chose it as my method.
- Write the manuscript by Latex and draw pictures with Visio and Python.

A method for optimizing the layout of sensors in a graph neural network water supply network 2023.09-2024.01

Published an **Invention Patent** as the **first author**, *National Natural Science Foundation of China (NSFC)*

Use graph convolutional network clustering algorithm to partition the water supply network into sub-regions, and arrange sensors in each region after partitioning, in order to better detect the entire water supply network system.

Volunteer for the 3rd Youth Academic Seminar of the Cross Science Department of the ***National Natural Science***

Foundation of China 2023.10

The 6th/7th **Academic Conference on Intelligent Optimization and Scheduling** *Wall Exhibition* 2024/2023.05

The 15th Guangdong **Hong Kong Macao Greater Bay Area Doctoral Student** Nanshan **Academic Forum and Tsinghua University** 706th **Doctoral Student Academic Forum** *Wall Exhibition* 2023.06

AI Speech Recognition Research 2020.04-2021-04

Leader *of College Students' Innovative Entrepreneurial Training Program, Beijing Municipal Level*

- Building and training a Chinese tone recognition model and published a simple speech recognition mini program.
- On the PyTorch platform, we extracted tone information features from the Tsinghua 30h dataset, and established a CNN+CTC model through VScode's remote SSH to connect remote service to use **GPU** for tone recognition training.
- Developed a speech recognition mini Wechat program by learning open-source code.
- Gained knowledge in areas such as Speech Signal Processing, Fourier Transform, Neural Networks, and Mini program development

WeChat Personalized Signature Mining and Word Cloud Visualization 2020.01-2020.02

Use Python, WordCloud, Jieba and Ichat to tokenize and visualize my **1000+** WeChat friends' personalized signature in a custom mask.

Taobao Double Eleven Big Data Analysis and Prediction 2023.04-2023.06

Various typical operations are involved in the entire data processing process of data preprocessing, storage, querying, and visualization analysis of **50 million** Taobao data in the Linux system.

- Familiar with importing and exporting data between different types of databases, such as using Sqoop to import Hive data into MySQL.
- Familiar with various Linux system operations, Vim editor-related operations, and MySql related operations.
- Building dynamic web applications using Eclipse involving technologies such as Linux, MySQL, Hadoop, Hive, Sqoop, Eclipse, ECharts and Spark.

English Learning System Based on Virtual Reality 2022.01-2022.04

Undergraduate graduation project

Use C#, SteamVr, Unity, VRTK and Vive Pro to build an English learning system.

Can be visited by https://youtu.be/-ZioAZY1JSs

HONORS & AWARDS

- A member of **Baidu AI** Pinecone Talent Development Elite Class (2024)
- Successfully passed the **WPS office C++ back-end** school-enterprise cooperation training camp (2024)
- **University of California Berkeley Summer Visit:** Learning Data Structures and Algorithms CS61b Course in EECS and Marketing course in Haas (2023)
- **Academic Scholarship (First Class) for three consecutive years** (2022-2024)
- High School **Information Technology Teaching Qualification Certificate** (2023)
- Summer School of **Peking University,** Elective Course on **Intelligent Computing Systems, Score 93** (2022)
- Speaker on behalf of the incoming graduate student (2022)
- **Head** of the **Academic Department of the Graduate Association** of the School of Computer Science (2022)
- Asia Pacific Cup Mathematical Modeling Success Award (2022)
- Third Prize at the 33rd Science and Technology Paper Report Conference (Title: Machine Learning based watershed runoff ensemble prediction and analysis) (2022)
- S Award in the **American Mathematical Modeling Competition** for College Students (2021)
- Several other awards, such as outstanding youth league cadres at the school level and excellent students in the three aspects of the school

PERSONAL SUMMARY

- I enjoy technical communication and often browse and learn on websites such as Blogs, Zhihu, Github, Bilibili.
- Passionate about learning, honest and upright, with the ability to conduct independent research.
- Love life and sports, enjoy dancing, humble and eager to learn, diligent and hardworking, with strong self-learning ability and ambition, as well as strong teamwork and communication skills.